



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE  
APPLICATIONS AND GEO-INFORMATICS**

**WEEKLY PROGRESS REPORT (26/02/2026 – 04/03/2026)**

**WEEK 9**

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| <b>PROJECT NAME</b> | <b>Monitoring and change detection system using OSINT</b> |
|---------------------|-----------------------------------------------------------|

**PROJECT DESCRIPTION :**

**CREATING OSINT TOOL FOR MONITORING AND  
CHANGE DETECTION SYSTEM BY MONITORING  
PUBLICLY AVAILABLE INFORMATION.**

**GROUP MEMBER :**

**Harshil Vaniya, Shalomo Agarwarka, Vrushank Thakkar**

**GROUP ID :**

**2026G195**

**GROUP GUIDE :**

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**DEPT. OF BIOCHEMISTRY AND FORENSICS SCIENCE, GUJARAT  
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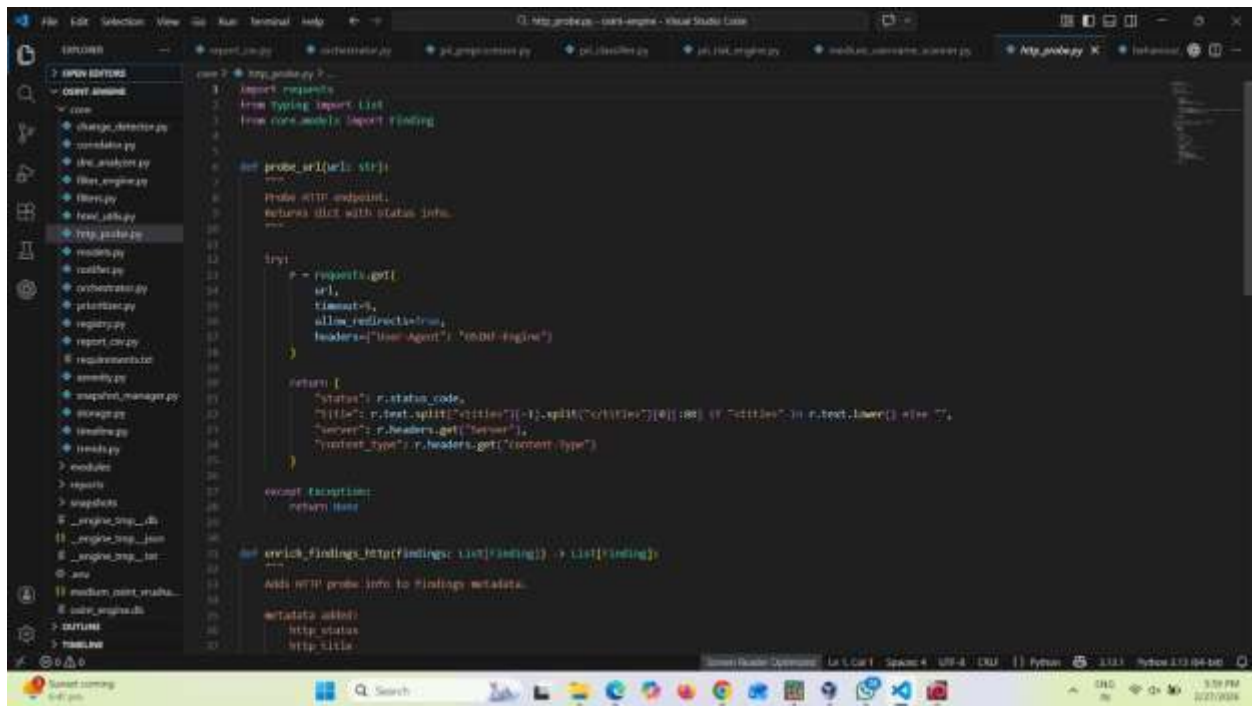
|                   |                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>26/02/2026</b> | <ul style="list-style-type: none"> <li>• Took a full day leave (Dissertation-related Presentations)</li> </ul>                                                                                                                                               |
| <b>27/02/2026</b> | <ul style="list-style-type: none"> <li>• error solving for cross-platform compatibility, but not solved yet (Vrushank Thakkar, Harshil Vaniya)</li> <li>• Solved download symbol issue for memory forensics in dissertation. (Shalomo Agarwarkar)</li> </ul> |
| <b>28/02/2026</b> | <b>Holiday (Saturday)</b>                                                                                                                                                                                                                                    |
| <b>1/03/2026</b>  | <b>Holiday (Sunday)</b>                                                                                                                                                                                                                                      |
| <b>2/03/2026</b>  | <ul style="list-style-type: none"> <li>• Performed error solving in cross-platform compatibility (Harshil Vaniya, Vrushank Thakkar)</li> <li>• Checked Maltego module dependencies. (Shalomo Agarwarkar)</li> </ul>                                          |
| <b>3/03/2026</b>  | <ul style="list-style-type: none"> <li>• Tested the tool with patches in Ubuntu. (Shalomo Agarwarkar)</li> <li>• only-live error solved and testing. (Harshil Vaniya, Vrushank Thakkar)</li> </ul>                                                           |
| <b>4/03/2026</b>  | <ul style="list-style-type: none"> <li>• Dhuleti Holiday</li> </ul>                                                                                                                                                                                          |

|                       |                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Next Week Plan</b> | <b>Integrating other OSINT-related modules and starting to work on GUI integration into the basic framework of tool</b> |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------|

## REFERENCE:

- <https://support.maltego.com/en/support/solutions/articles/15000053545-building-integrations-for-maltego>
- <https://support.maltego.com/en/support/solutions/articles/15000017584-setup-python-transform-server-using-trx-gunicorn>
- <https://runebook.dev/en/docs/python/using/unix/python-related-paths-and-files>
- <https://medium.com/@sainath.814/how-to-fix-and-troubleshoot-dns-issues-in-linux-step-by-step-guide-d69a3ca42f7a>

## SCREEN SHOTS:



```
1 import requests
2 from typing import List
3 from core.models import Finding
4
5
6 def probe_url(url: str):
7     """
8     Probe HTTP endpoint.
9     Returns dict with status info.
10    """
11
12    try:
13        r = requests.get(
14            url,
15            timeout=5,
16            allow_redirects=True,
17            headers={"User-Agent": "m00t-engine"}
18        )
19
20        return {
21            "status": r.status_code,
22            "title": r.text.split("title=")[-1].split("</title>")[0] if "<title>" in r.text.lower() else "",
23            "server": r.headers.get("server"),
24            "content_type": r.headers.get("content-type")
25        }
26    except Exception:
27        return None
28
29
30 def enrich_findings_http(findings: List[Finding]) -> List[Finding]:
31     """
32     Adds HTTP probe info to findings metadata.
33    """
34
35    metadata added:
36    http status
37    http title
```

27-02-26 Shalomo Agarwarkar : Solved the download speed issue in dessertation

```
Windows PowerShell
[!] Downloading PDB file...
[*] Downloading PDB from: https://msdl.microsoft.com/download/symbols/ntkrnlmp.pdb/87A00A457156E2568E5C7F2E20A2429D1/ntkrnlmp.pdb
Saving to: "symbols\\87A00A457156E2568E5C7F2E20A2429D1\\ntkrnlmp.pdb"
[*] Fetching remote file size...
[!] Could not get file size, falling back to single connection download...
[-] Download failed
[!] PDB download failed!

=====
PERFORMANCE METRICS
=====

[1] EXECUTION TIMING
Wall Time:          48.47 sec
CPU User Time:      2.69 sec
CPU Kernel Time:    2.69 sec
CPU Total Time:     5.38 sec
Avg CPU Utilization: 0.92 %%

[2] PEAK RESOURCE CONSUMPTION
Peak Working Set (RSS): 25.80 MB
Peak Private Commit: 21.71 MB
Peak Handle Count: 192
Peak Thread Count: 8

=====
Exit Code:
=====
PS D:\Dessertation\RAM analysis\cmake-build-debug>
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\Dessertation\RAM analysis\cmake-build-debug> .\1.ps1
=====
RAM ANALYSIS PERFORMANCE & RESOURCE MONITOR
=====

Launching RAM Analysis...

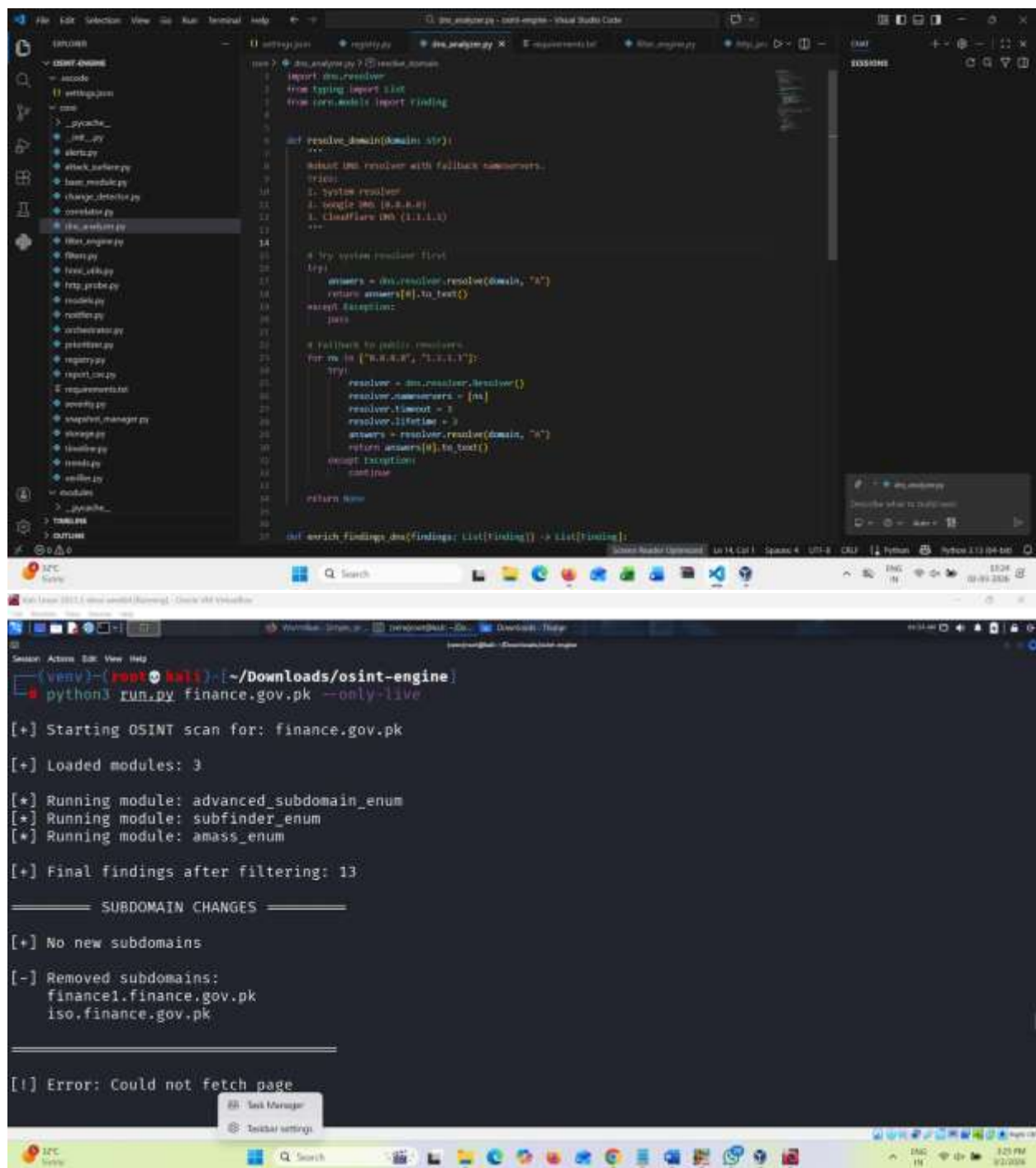
Enter memory dump file path: D:\RAM\ramdump.raw

[*] Finding DTB...
[+] DTB found: 0x1ae000

[*] Finding Kernel Base...
[*] KDBG method failed, trying SystemRoot method...
[+] Kernel Base: 0xf80782600000

[*] Scanning kernel PE for PDB signature...
[+] Successfully extracted PDB information
GUID: 87A00A457156E2568E5C7F2E20A2429D
Age: 1
PDB: ntkrnlmp.pdb
Download URL: https://msdl.microsoft.com/download/symbols/ntkrnlmp.pdb/87A00A457156E2568E5C7F2E20A2429D1/ntkrnlmp.pdb

[*] Downloading PDB file...
[*] Downloading PDB from: https://msdl.microsoft.com/download/symbols/ntkrnlmp.pdb/87A00A457156E2568E5C7F2E20A2429D1/ntkrnlmp.pdb
Saving to: "symbols\\87A00A457156E2568E5C7F2E20A2429D1\\ntkrnlmp.pdb"
[*] Fetching remote file size...
[!] Could not get file size, falling back to single connection download...
[-] Download failed
[!] PDB download failed!
```



Browser tabs: WhatsApp, Screenshots - Google Docs, maltego module creation - Search, Maltego Building Integrations for - , New tab

Address bar: <https://support.maltego.com/en/support/solutions/articles/15000053545-building-integrations-for-maltego>

Navigation: Home, Knowledge base, Login English

Find some solutions here...

### Introduction

This guide aims to walk you through the process of designing, planning, developing, and launching your own Maltego integration - keeping it as simple as possible to help you get started quickly.

Whether you're an amateur OSINT enthusiast, a professional Maltego user who needs to access internal data and systems within Maltego, or a data provider looking to build a commercial integration for the Data Hub, this guide is here to help you get started.

Integrations are what make Maltego so useful. Maltego is a highly flexible tool and can be customized to display and interface with almost any API-driven system you can think of. Doing so is often easier than people think. Writing Transforms is quite simple—the difficulties usually lie in understanding how a Maltego integration works and is set up, and then in designing useful and reusable Transforms. This guide aims to provide brief, effective, and easy-to-follow guidance, especially in these areas.

### Why a Maltego Integration?

If you're here, chances are you've already decided you want to write an integration for Maltego. Whether that's the case or not, here are some of the main reasons and benefits of Maltego integrations:

- **Visualize and traverse your data.** Maltego can be thought of as a generic frontend: it adds an extra interface for data or tools you already work with and helps you visually explore them.
- **De-silo your data.** A good Maltego integration makes disparate systems seamlessly compatible in one user interface.
- **Streamline investigations.** Jump between data points, spot patterns, and connect the dots in ways that a spreadsheet and text document just don't allow.

It's easy. Don't be intimidated—this guide is lengthy because we spend a fair amount of time talking about processes and best practices. Writing the integration code is quite straightforward.

### Cheat Sheet

This guide will teach you what you need to know, but if you want a quick reference (and checklist) to refer back to later or throughout the process, use the **Cheat Sheet**.

### Articles in this folder -

- Writing Transforms
- Building Integrations for Maltego
- Building Maltego Integrations - Cheat Sheet
- Transform Libraries
- Development Transform Server
- Production Transform Server using Gunicorn
- Setup: Python Transform Server using TRX + Gunicorn
- Python Transform Server using TRX + Apache2
- Production Transform Server using Apache2
- Set up the TDS
- Add Maltego-TRX Transforms to Maltego Desktop Client via ITDS
- Local Transforms Example

Taskbar: 37°C Sunny, Search, 16:11 03-03-2026

OSINT Engine Output:

```
root@blue:/home/vboxuser/Downloads/osint-engine

===== TREND ANALYSIS =====

Target: finance.gov.pk

Subdomain volatility : MEDIUM
Growth rate          : 0.0
Stability score       : 0.84

===== CURRENT FINDINGS =====

Finding(type='subdomain', value='api-economy.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397759+00:00', metadata={'sources': ['hackertarget'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 404, 'title': '', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'application/json; charset=utf-8'}, 'verified': True})
Finding(type='subdomain', value='bak-economy.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397791+00:00', metadata={'sources': ['hackertarget'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 200, 'title': 'pakistan economy dashboard - admin panel</title>', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'text/html'}, 'verified': True})
Finding(type='subdomain', value='economy.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397798+00:00', metadata={'sources': ['hackertarget'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 200, 'title': 'pakistan economy dashboard</title>', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'text/html'}, 'verified': True})
Finding(type='subdomain', value='pub-economy.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397806+00:00', metadata={'sources': ['hackertarget'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 404, 'title': '', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'application/json; charset=utf-8'}, 'verified': True})
Finding(type='subdomain', value='mail.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397802+00:00', metadata={'sources': ['hackertarget'], 'ip': '210.56.14.219', 'resolves': True, 'http': {'alive': False, 'verified': False}})
Finding(type='subdomain', value='webmail.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397810+00:00', metadata={'sources': ['hackertarget'], 'ip': '210.56.14.219', 'resolves': True, 'http': {'alive': False, 'verified': False}})
Finding(type='subdomain', value='ftp.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.99, timestamp='2026-03-03T06:57:49.397819+00:00', metadata={'sources': ['dns_bruteforce'], 'ip': '210.56.14.219', 'resolves': True, 'http': {'alive': False, 'verified': False}})
Finding(type='subdomain', value='www.finance.gov.pk', source='hackertarget', target='finance.gov.pk', confidence=0.85, timestamp='2026-03-03T06:57:49.397814+00:00', metadata={'sources': ['hackertarget'], 'ip': '210.56.14.219', 'resolves': True, 'http': {'alive': False, 'verified': False}})

[+] Email sent successfully
```

Taskbar: Watchlist Ideas, Search, 12:28 03-03-2026

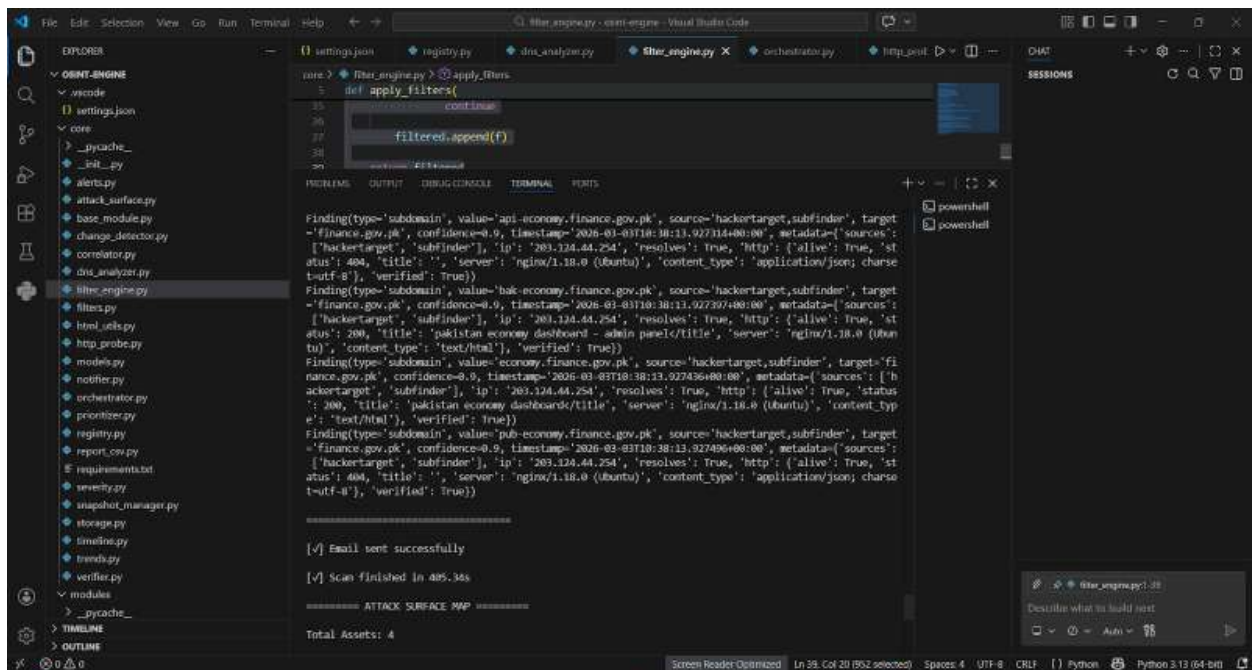


```

root@blue:/home/ktaxuser/Downloads/ecint-engine#
[+] Scan finished in 44.93s
===== ATTACK SURFACE MAP =====
Total Assets: 8
INFRASTRUCTURE GROUPING (BY IP)
203.124.64.234
├── api-economy.finance.gov.pk
├── bak-economy.finance.gov.pk
├── economy.finance.gov.pk
├── pub-economy.finance.gov.pk
218.54.14.219
├── mail.finance.gov.pk
├── webmail.finance.gov.pk
├── ftp.finance.gov.pk
├── www.finance.gov.pk
TECH STACK DISTRIBUTION
nginx/1.18.8 (Ubuntu)
├── api-economy.finance.gov.pk
├── bak-economy.finance.gov.pk
├── economy.finance.gov.pk
├── pub-economy.finance.gov.pk
HIGH RISK ASSETS
None detected
[+] Scan complete, found 8 findings.
(wen) root@blue:/home/ktaxuser/Downloads/ecint-engine#

root@blue:/home/ktaxuser/Downloads/ecint-engine#
[+] Loaded modules: 3
[*] Running module: advanced_subdomain_enum
[*] Running module: subfinder_enum
[*] Running module: anssl_enum
[+] Final findings after filtering: 8
===== SUBDOMAIN CHANGES =====
[+] No new subdomains
[-] Removed subdomains:
cms.finance.gov.pk
lists.cms.finance.gov.pk
apc.finance.gov.pk
pran.finance.gov.pk
adm.finance.gov.pk
server.finance.gov.pk
[+] Error: Could not fetch page
===== SEVERITY REPORT =====
[MEDIUM] removed_subdomain -> cms.finance.gov.pk
[MEDIUM] removed_subdomain -> lists.cms.finance.gov.pk
[MEDIUM] removed_subdomain -> apc.finance.gov.pk
[MEDIUM] removed_subdomain -> pran.finance.gov.pk
[MEDIUM] removed_subdomain -> adm.finance.gov.pk
[MEDIUM] removed_subdomain -> server.finance.gov.pk
===== ALERTS =====
[CRITICAL] SUBDOMAIN_REMOVED -> ['cms.finance.gov.pk', 'lists.cms.finance.gov.pk', 'apc.finance.gov.pk', 'pran.finance.gov.pk', 'adm.finance.gov.pk', 'server.finance.gov.pk']
[MEDIUM] VOLATILITY_INCREASE -> Moderate subdomain churn detected

```



```
def apply_filters(finding):
    # Filter findings based on various criteria
    # ... (code omitted) ...
    filtered.append(finding)

# Example findings
findings = [
    Finding(type='subdomain', value='api-economy.finance.gov.pk', source='hackertarget', subfinder='hackertarget', target='finance.gov.pk', confidence=0.9, timestamp='2026-03-03T10:38:13.927397400:00', metadata={'sources': ['hackertarget', 'subfinder'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 404, 'title': ''}, 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'application/json; charset=utf-8'}, 'verified': True)),
    Finding(type='subdomain', value='bak-economy.finance.gov.pk', source='hackertarget', subfinder='hackertarget', target='finance.gov.pk', confidence=0.9, timestamp='2026-03-03T10:38:13.927397400:00', metadata={'sources': ['hackertarget', 'subfinder'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 200, 'title': 'Pakistan economy dashboard - admin panel', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'text/html'}, 'verified': True)),
    Finding(type='subdomain', value='economy.finance.gov.pk', source='hackertarget', subfinder='hackertarget', target='finance.gov.pk', confidence=0.9, timestamp='2026-03-03T10:38:13.927426400:00', metadata={'sources': ['hackertarget', 'subfinder'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 200, 'title': 'Pakistan economy dashboard', 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'text/html'}, 'verified': True)),
    Finding(type='subdomain', value='pub-economy.finance.gov.pk', source='hackertarget', subfinder='hackertarget', target='finance.gov.pk', confidence=0.9, timestamp='2026-03-03T10:38:13.927496400:00', metadata={'sources': ['hackertarget', 'subfinder'], 'ip': '203.124.44.254', 'resolves': True, 'http': {'alive': True, 'status': 404, 'title': ''}, 'server': 'nginx/1.18.0 (Ubuntu)', 'content_type': 'application/json; charset=utf-8'}, 'verified': True))
]

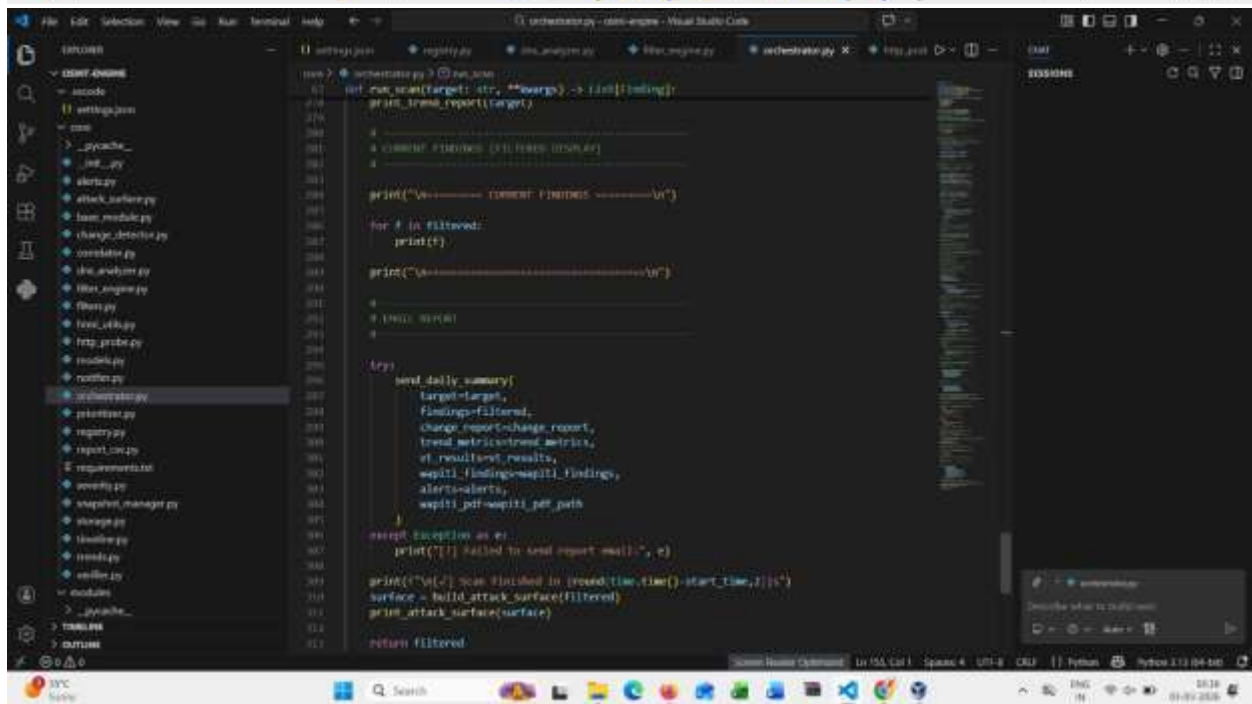
# Apply filters
filtered = apply_filters(finding)

# Output
print([f for f in filtered])

# Email sent successfully
print([f for f in filtered])

# Scan finished in 405.34s

# ===== ATTACK SURFACE MAP =====
# Total Assets: 4
```



```
def run_scan(target: str, **kwargs) -> dict[dict[dict]]:
    # Run scan on target
    # ... (code omitted) ...
    # Filter findings
    filtered = apply_filters(finding)

    # Generate report
    report = generate_report(filtered)

    # Send alert
    send_alert(filtered)

    # Send daily summary
    send_daily_summary(target=target, findings=filtered, change_report=change_report, trend_metrics=trend_metrics, vt_results=vt_results, wapiti_findings=wapiti_findings, alerts=alerts, wapiti_pdf=wapiti_pdf_path)

    # Return filtered findings
    return filtered
```



```
Kali Linux 2017.2 virtual machine (Running) - Oracle VM VirtualBox
Session: Actions Edit View Help
[vmx] root@kali: ~/Downloads/OSINT-Monitoring-and-Change-Detection-System-main
python3 run.py finance.gov.pk --only-live

[+] Starting OSINT scan for: finance.gov.pk
[+] Loaded modules: 3
[*] Running module: advanced_subdomain_enum
[*] Running module: subfinder_enum
[*] Running module: amass_enum
[+] Final findings after filtering: 0

----- SUBDOMAIN CHANGES -----
[+] No new subdomains
[-] No removed subdomains

----- SEVERITY REPORT -----
[v] No alerts triggered
```

```
Kali Linux 2017.2 virtual machine (Running) - Oracle VM VirtualBox
Session: Actions Edit View Help
[v] Email sent successfully
[v] Scan finished in 320.9s

----- ATTACK SURFACE MAP -----
Total Assets: 0
INFRASTRUCTURE GROUPING (BY IP)
No IP mapping available
TECH STACK DISTRIBUTION
No server fingerprinting data
HIGH RISK ASSETS
None detected

[+] Scan complete. Found 0 findings.

[vmx] root@kali: ~/Downloads/OSINT-Monitoring-and-Change-Detection-System-main
python3 run.py finance.gov.pk --only-verified
```